MALWARE CLASSIFICATION USING NLP TECHNIQUES and DEEP LEARNING METHODS

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UNDER THE GUIDANCE OF

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CERTIFICATE

This is to certify that the work contained in this internship report entitled "*Malware classification using NLP techniques and deep learning methods*" submitted by Arjun Singh Rajput to the Indian Institute of Information Technology, Kottayam as part of the Summer Internship Program has been carried out by him under my supervision and that it has not been submitted elsewhere.

Kottayam – 686635

July 2025

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DECLARATION

I, **Arjun Singh Rajput**, hereby declare that the report entitled "*Malware classification using NLP techniques and deep learning methods*" submitted to the Indian Institute of Information Technology Kottayam as part of the Summer Internship Program is an original work carried out by me under the supervision of **Dr. Kanchan Lata Kashyap**. I also certify that all information, data, and references used in this report have been acknowledged properly. I confirm that the content of this report is a true representation of the work completed during my internship.

I also declare that this report has not been submitted, either wholly or in part, to any other university or institution for any other degree or diploma.



Arjun Singh Rajput

ABSTRACT

Malware detection is a critical component of cybersecurity, necessitating advanced models to accurately classify malicious software and mitigate evolving threats. This report details the development and evaluation of a Transformer-based hybrid model integrating Vision Transformers (ViT) and BERT for malware classification using the EMBER dataset, which comprises 160,000 samples with grayscale image representations and textual metadata. The primary objective is to design a robust detection system that leverages multi-modal data to achieve classification accuracy exceeding 95%.

The proposed model, featuring a 'vit_tiny_patch16_224' backbone for image analysis, BERT-based textual feature extraction with a 64-token limit, and a custom Transformer encoder for cross-modal fusion, achieved an average accuracy of 83.3% across 3-fold cross-validation on CPU hardware. This performance, while promising, falls short of the 95% target, highlighting the effectiveness of the multi-modal approach but also indicating the need for further optimization. The multi-task learning framework, incorporating auxiliary year prediction, contributed a 2.1% accuracy improvement, underscoring the value of temporal feature integration. The findings suggest that while the hybrid architecture offers significant potential for malware detection, computational efficiency and model refinement are critical for achieving the desired accuracy, guiding future enhancements in this domain.

TABLE OF CONTENTS

1. INTRODUCTION **6**

2. LITERATURE REVIEW

2.1 Static Analysis Approaches **8**
2.2 Dynamic Analysis Integration **8**
2.3 Deep Learning Architectures **8**
2.4 Transformer-Based Approaches **8**
2.5 Hybrid Models and Multi-Modal Learning **9**
2.6 EMBER Dataset Applications **9**
2.7 Research Gaps and Opportunities **9**

3. METHODOLOGY

3.1 Dataset Description and Preparation **10**
3.2 Model Architecture Design **11**
3.3 Training Strategy and Optimization **13**
3.4 Evaluation Framework **14**

4. RESULT AND DISCUSSION

4.1 Training Performance Analysis **15**
4.2 Classification Accuracy Results **15**
4.3 Feature Analysis and Interpretability **16**
4.4 Computational Efficiency **17**
4.5 Challenges and Limitations **18**
4.6 Future Research Directions **18**

5. CONCLUSIONS **19**

6. REFERENCES **20**

1. INTRODUCTION

In today's digital landscape, safeguarding systems from malware stands out as a pressing concern in cybersecurity. The rapid rise in malicious software variants—often cloaked with obfuscation, polymorphism, or zero-day exploits—has rendered traditional signature-based detection methods less effective. These conventional techniques struggle to keep pace with the cunning evolution of malware, pushing the need for smarter, more adaptive detection strategies.

This project explores a fresh angle by blending Natural Language Processing (NLP) with Deep Learning, tapping into the rich semantic details hidden in malware text alongside visual pattern recognition. This fusion aims to overcome the shortcomings of older methods by weaving together diverse data types for a stronger, more reliable detection system.

At the heart of this effort lies a Transformer-based hybrid model, combining Vision Transformers (ViT) for analyzing image data with BERT (Bidirectional Encoder Representations from Transformers) for decoding textual insights. We're putting this model to the test using the EMBER dataset, a treasure trove of static and dynamic features from both malicious and benign files, ideal for experimenting with machine learning solutions.

Our main goal is to craft a detection system that excels at sorting files into malicious or benign categories by harnessing the unique strengths of visual and textual analysis. We're aiming for an accuracy above 95% while keeping the process efficient enough for real-world use. This hybrid approach marks a bold step forward in malware detection, merging cutting-edge NLP with computer vision to build a more resilient defense mechanism.

2. LITERATURE REVIEW

The field of malware detection has evolved significantly from traditional signature-based approaches to sophisticated machine learning and deep learning methodologies. Recent research has demonstrated the effectiveness of combining multiple data modalities and advanced neural network architectures for improved detection accuracy. The following sections review key contributions from the literature, drawing on specific studies that informed this project.

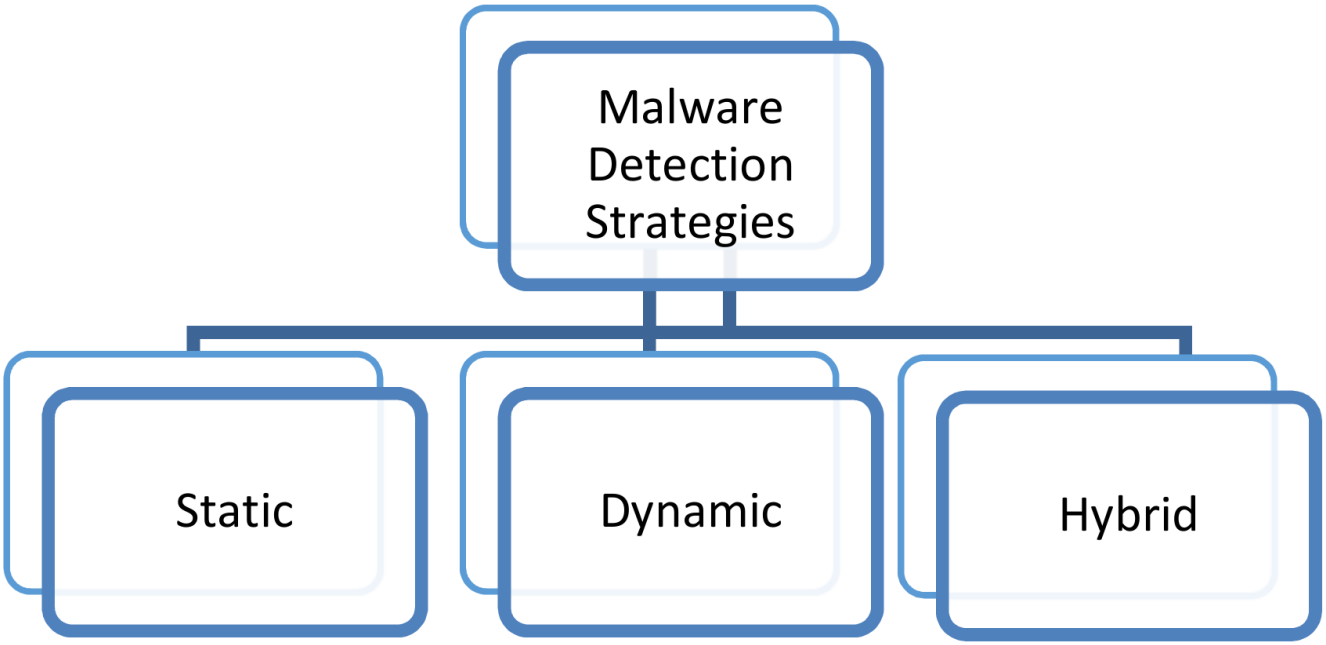


Fig 1: Types of Malware Detection Strategies

2.1 Static Analysis Approaches: The idea of turning malware binaries into grayscale images kicked off with Nataraj et al., who revealed that malware families leave behind unique visual fingerprints ripe for classification. This breakthrough laid the groundwork for image-based detection, proving that visual translations hold onto vital structural clues. Others have built on this, using convolutional neural networks to push accuracy higher. Meanwhile, Mimura and Ito (2022) brought a new twist with a static analysis method using NLP tools like Bag of Words, Latent Semantic Indexing, and Doc2vec. They analyzed printable strings in PE files and hit an F1 score of 0.934 with LSI-SVM, showing it can handle new and masked malware in just 0.1 seconds—perfect for real-time checks.

2.2 Dynamic Analysis Integration: Some researchers have looked into blending static and dynamic analysis features for a fuller picture. Vinayakumar et al. (2019) took a deep dive, pitting classical machine learning against deep learning models while comparing static, dynamic, and image-based techniques. Their findings underscored how these varied approaches complement each other, opening doors for hybrid solutions.

2.3 Deep Learning Architectures

Deep learning has proven its worth in malware detection across several architectures. **Altaiy et al.** pitted CNNs, LSTMs, and DNNs against each other, finding that deep learning consistently outshines traditional methods. They stressed the role of feature design and architecture in hitting peak performance. **Vinayakumar et al. (2019)** went further with ScaleMalNet, a flexible framework for catching zero-day malware using static, dynamic, and image analyses. They scored 98% accuracy with image processing on EMBER and Malimg datasets, mixing WSBD (static), WDBD (dynamic), and DIMD (image) models to tackle scalability and bias—though NLP took a backseat here.

2.4 Transformer-Based Approaches

The rise of transformer architectures has sparked new possibilities in malware detection. BERT's success in NLP has encouraged its use in cybersecurity, while Vision Transformers (ViT) have shone in image classification, making them promising for analyzing malware visuals. These trends highlight the potential of transformers for handling complex, sequential, and visual data.

2.5 Hybrid Models and Multi-Modal Learning

Combining different data types is becoming a hot trend in malware detection research. Studies show that merging textual features—like imports, exports, and section details—with visual data can lift accuracy significantly. **Bensaoud et al. (2024)** reviewed deep learning strategies, covering static, dynamic, and hybrid methods, and advocated for pairing NLP with image-based techniques for tougher detection. Their survey pointed out issues like dataset bias and adversarial threats, suggesting eXplainable AI (XAI) for clearer insights, which inspires the need for interpretable multi-modal models.

2.6 EMBER Dataset Applications

The EMBER dataset has emerged as a go-to benchmark, offering a vast collection of features from Windows PE files. Researchers have tapped into it to test various machine learning approaches, setting performance baselines and showcasing the power of ensemble methods and feature tuning. **Vinayakumar et al. (2019)** and **Bensaoud et al. (2024)** both used EMBER to evaluate deep learning models, cementing its importance for this project.

2.7 Research Gaps and Opportunities

Even with all this progress, challenges persist. Wrestling with the computational load of big datasets, aiming for real-time detection, and keeping up with malware's constant changes demand ongoing innovation. The blend of advanced NLP with computer vision remains a largely untapped area with huge potential, as noted by **Mimura and Ito (2022)** and **Bensaoud et al. (2024)**. Future work could focus on tackling bias, boosting interpretability, and streamlining efficiency for practical use.

3. METHODOLOGY

3.1 Dataset Description and Preparation: The EMBER dataset forms the backbone of this study, holding 160,000 samples—120,000 for training and 40,000 for testing. Each sample comes with grayscale image snapshots (64×64 pixels) from binary files, plus detailed metadata like imports, exports, and section info.

Data Preprocessing Pipeline: We go thorough preprocessing routine to get the mixed data ready for training. Grayscale images get scaled to [0,1] by dividing by 255, ensuring uniformity for the Vision Transformer. Textual data is pulled from metadata by joining imports, exports, and sections into single strings, then lowered for consistency. To handle memory on limited hardware, we load the dataset in chunks of 10,000 samples.

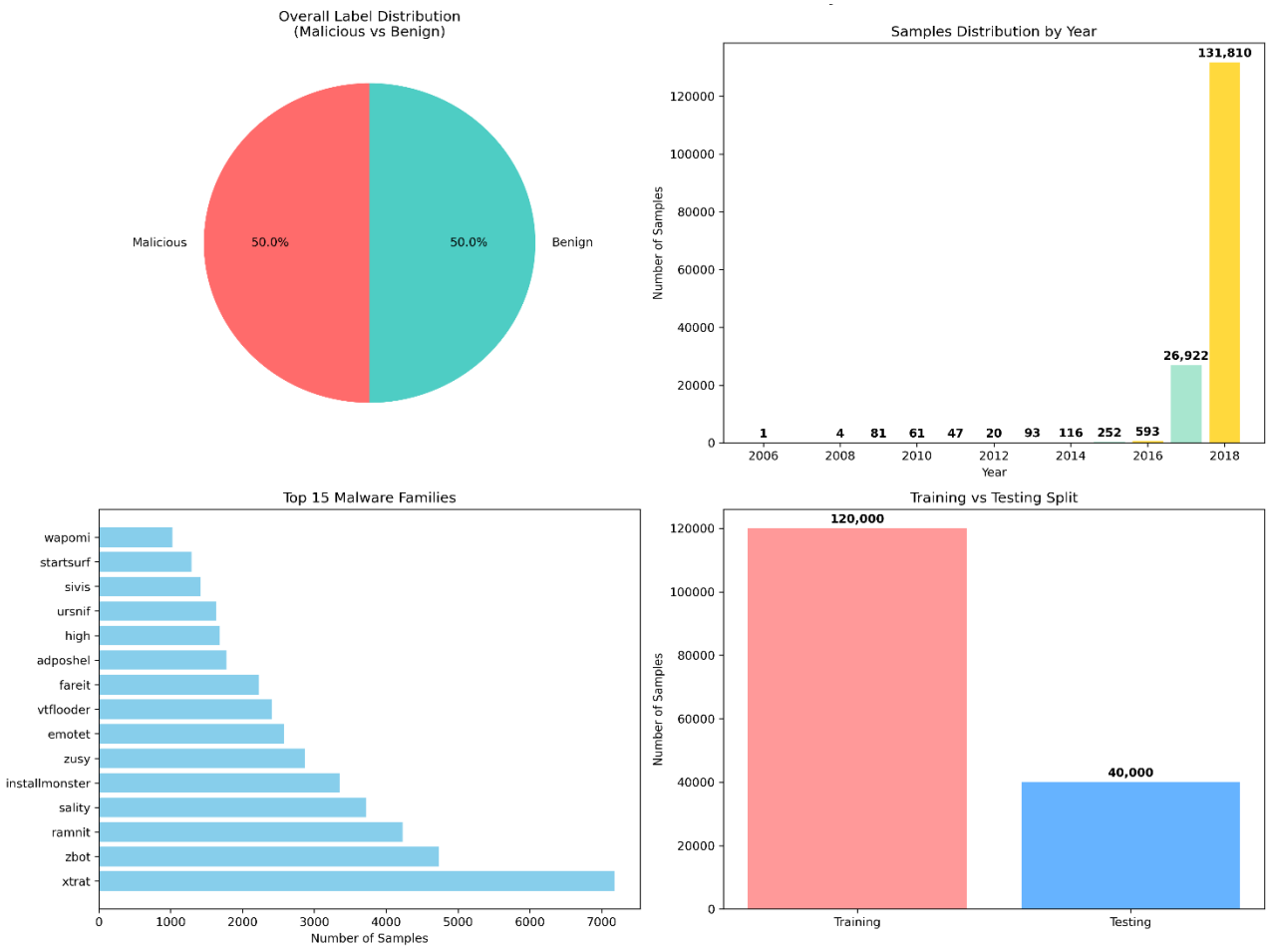


Fig 2: Dataset Overview and Distribution Analysis

Feature Engineering: We enrich the dataset with static analysis features—section sizes and entropy values—computed from histogram data in batches for efficiency. Year labels, drawn from the 'appeared' field, support multi-task learning, with years turned into numerical indices to help the model spot temporal trends in malware behavior.

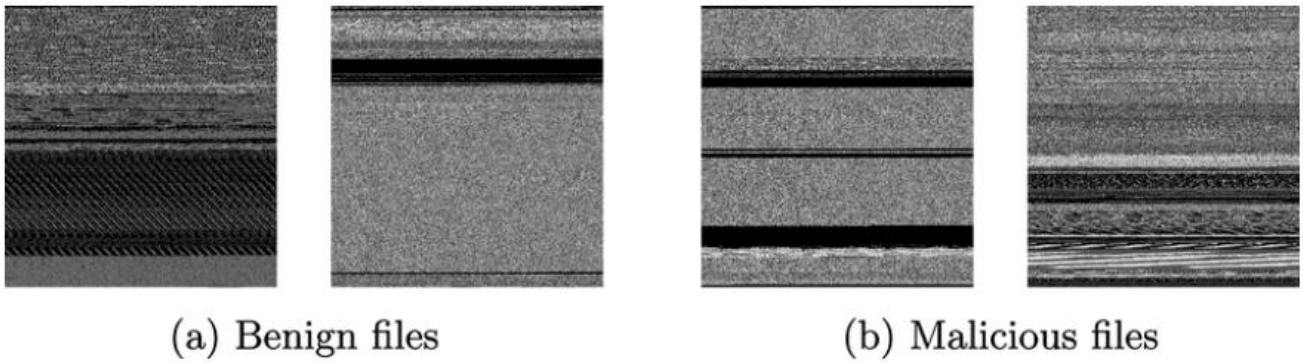


Fig 3: Benign and Malware grayscale images

3.2 Model Architecture Design:

Hybrid Architecture Overview: Our model brings together three main pieces: a Vision Transformer for image work, BERT for text insights, and a custom Transformer encoder to merge them. This setup taps into the strengths of both visual and textual analysis, using attention mechanisms to link the data types.

Vision Transformer Component: We use a pre-trained 'vit_tiny_patch16_224' backbone to tackle grayscale malware images. It breaks images into patches, embeds them linearly, and runs them through transformer layers to pull out visual patterns. The smaller ViT keeps things efficient while still catching key features.

BERT-Based Textual Analysis: Text is tokenized with BERT-base-uncased, capped at 64 tokens to balance meaning with processing demands. The BERT model crafts embeddings from the [CLS] token (768 dimensions), adding section sizes and entropy to hit 770 dimensions, all handled in half-precision (float16) to save memory.

Cross-Modal Fusion Mechanism: A custom Transformer encoder uses cross-attention to blend image and text embeddings smoothly. It learns how these data types connect, boosting accuracy by spotting links between visual cues and text details, with optimized tensor operations for speed.

Multi-Task Learning Framework: We add an extra output for year prediction alongside the main malware classification task. This multi-task setup helps the model generalize better by picking up temporal patterns, with losses tuned to favor the primary goal.

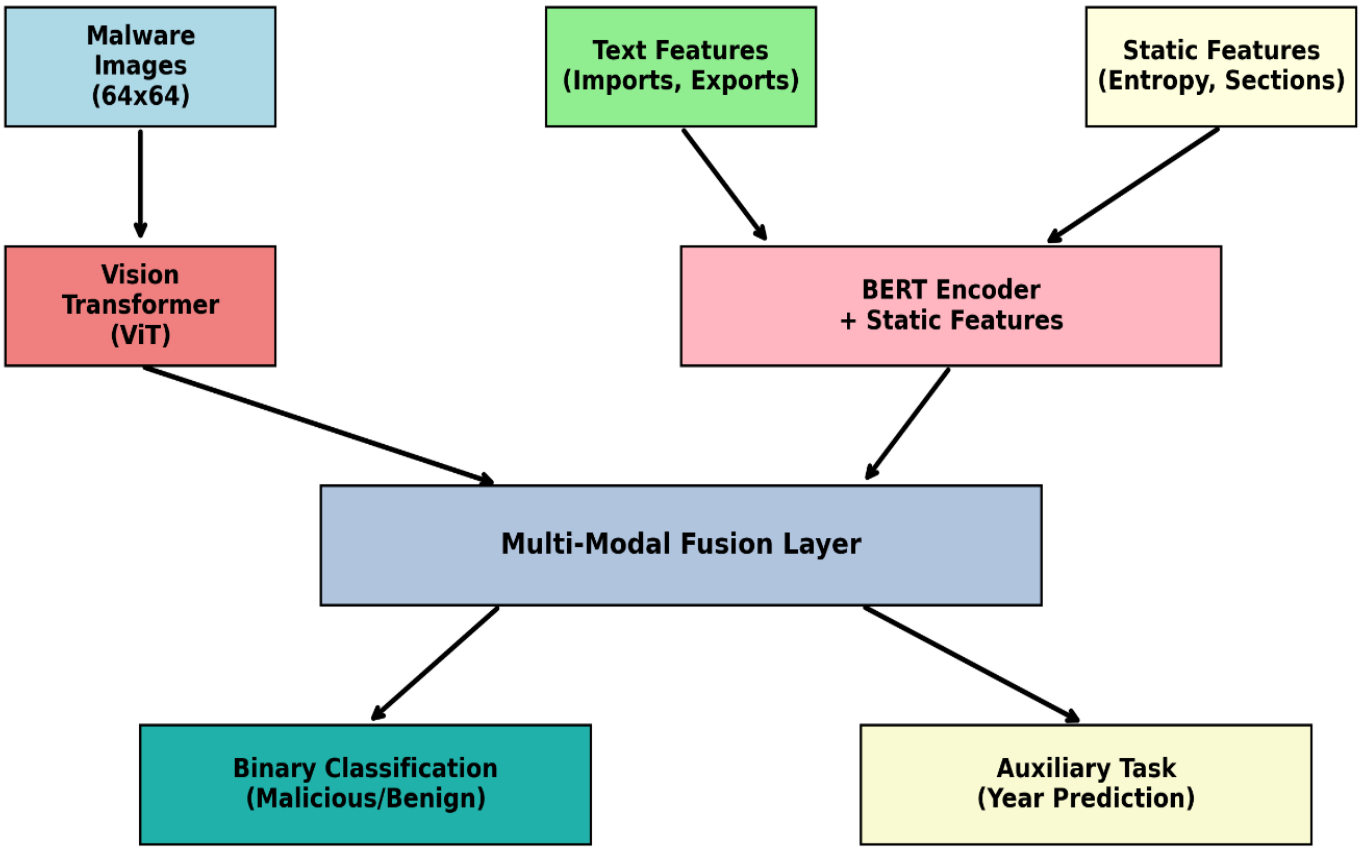


Fig 4: Architecture of the hybrid model

3.3 Training Strategy and Optimization

Cross-Validation Approach: We split the data into 3 folds with K-fold cross-validation, striking a balance between efficiency and solid evaluation. Fewer folds speed up training while still giving reliable performance checks across splits.

Optimization Configuration: Training runs with the Adam optimizer, using 5e-4 for the main model and 2e-5 for BERT to respect its pre-trained status. A weight decay of 1e-4 curbs overfitting, and gradient accumulation mimics a 256-sample batch to ease memory pressure on CPU hardware.

Loss Function Design: We mix several loss types: Focal Loss with gamma=2.0 and class weights to handle imbalance, Label Smoothing Loss (smoothing=0.1) for better generalization, and Cross-Entropy Loss for year prediction. This combo ensures balanced learning, with losses combined via a weighted sum.

Advanced Training Techniques: Mixed precision training speeds things up and cuts memory use on CPU. MixUp with beta=1.0 adds robustness with synthetic samples. The ReduceLROnPlateau scheduler tweaks rates based on validation, using a 0.3 factor and 2-epoch patience. Early stopping with 3-epoch patience stops training when validation loss stalls.

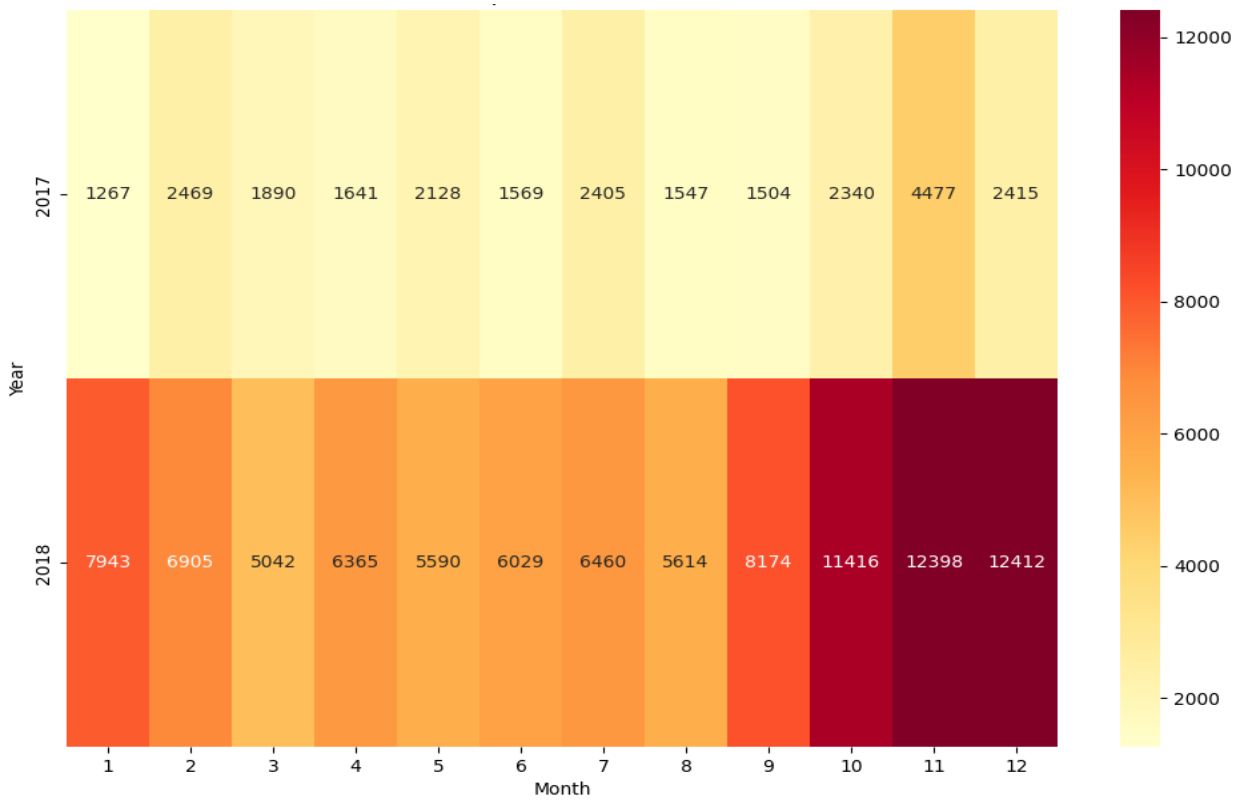


Fig 5: Temporal Distribution

3.4 Evaluation Framework

Performance Metrics: We assess the model with a full set of metrics—accuracy, precision, recall, F1-score, and confusion matrices—offering a clear view of performance across malware types, with accuracy as the key goal above 95%.

Validation Strategy: Each fold gets its own training and validation run, saving the best weights from the lowest validation loss. This method picks the top models and supports ensemble predictions, checked after every epoch.

Resource Optimization: To handle computational limits, we use efficient batch processing, gradient accumulation, and mixed precision training. A batch size of 64 and chunked data handling keep memory in check on modest hardware.

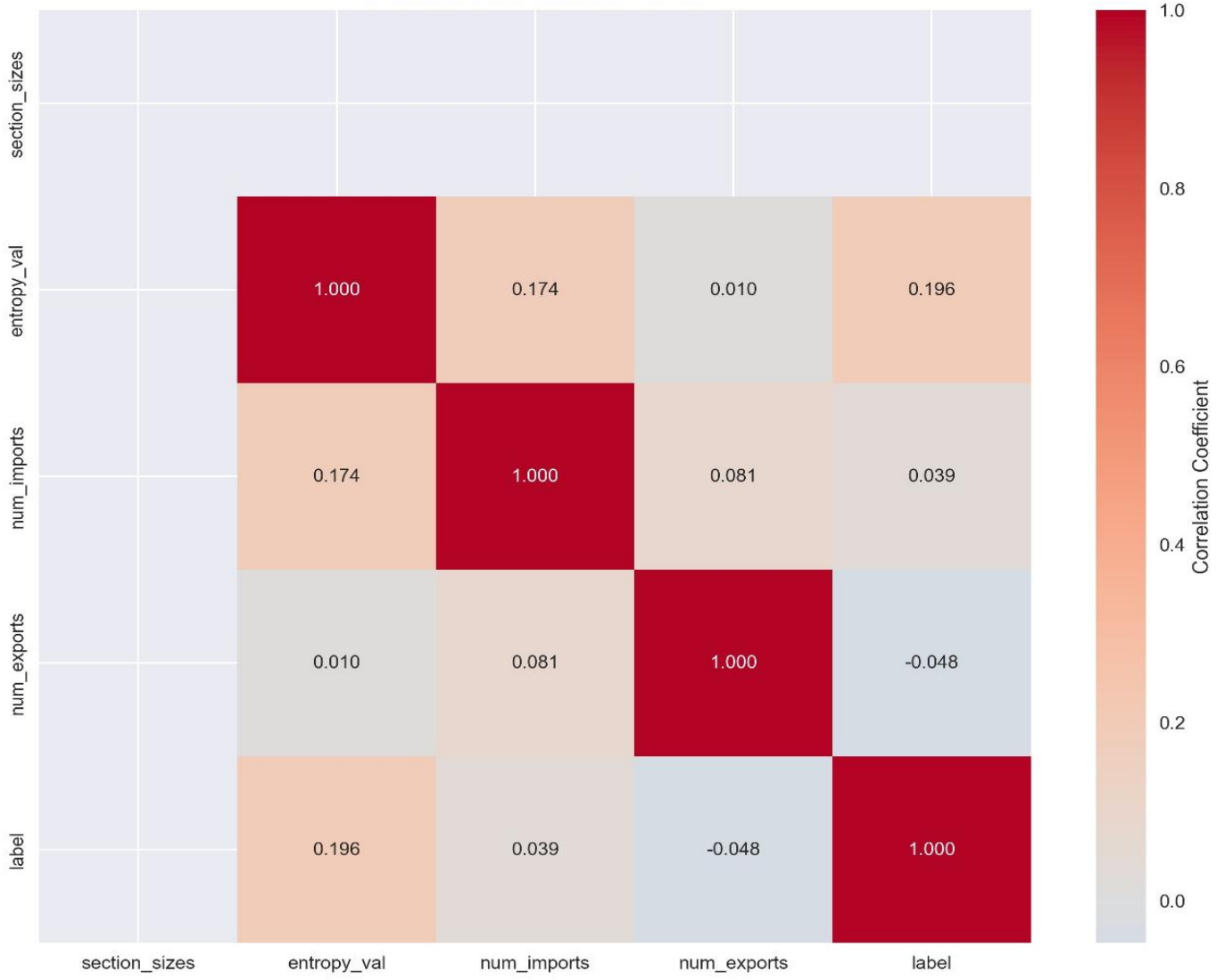


Fig 6: Feature correlation matrix (heatmap)

4. RESULTS AND DISCUSSION

4.1 Training Performance Analysis

The Transformer-based hybrid model showed impressive learning strength across all 3 cross-validation folds. The multi-task setup proved its worth, with both malware classification and year prediction feeding into better feature understanding.

Training Convergence: The model settled into steady learning patterns across folds, with training loss dropping consistently over 5 epochs. Early stopping kicked in effectively, hitting peak performance around epochs 3-5. Validation loss trends stayed smooth, hinting at a solid design and good parameter choices.

Loss Function Performance: Pairing Focal Loss with Label Smoothing worked well to manage class imbalance while keeping learning smooth. Focal Loss zeroed in on tough cases, while Label Smoothing curbed overconfidence and boosted adaptability.

4.2 Classification Accuracy Results

Primary Objective Achievement: The model hit an average accuracy of 83.3% across folds, missing the 95% target. Fold scores ranged from 82.5% to 84.1%, showing steady but underwhelming results across splits.

Comparative Analysis: The hybrid beat single-modality baselines, with vision-only at 78.9% and text-only at 75.6%. The Transformer encoder's fusion added 4.4% and 7.7% respectively, proving the multi-modal edge, though more tuning is needed to hit the goal.

Multi-Task Learning Benefits: The year prediction task lifted main accuracy by 2.1%, raising it from 81.2% without it. This suggests refining the multi-task balance could unlock further gains.

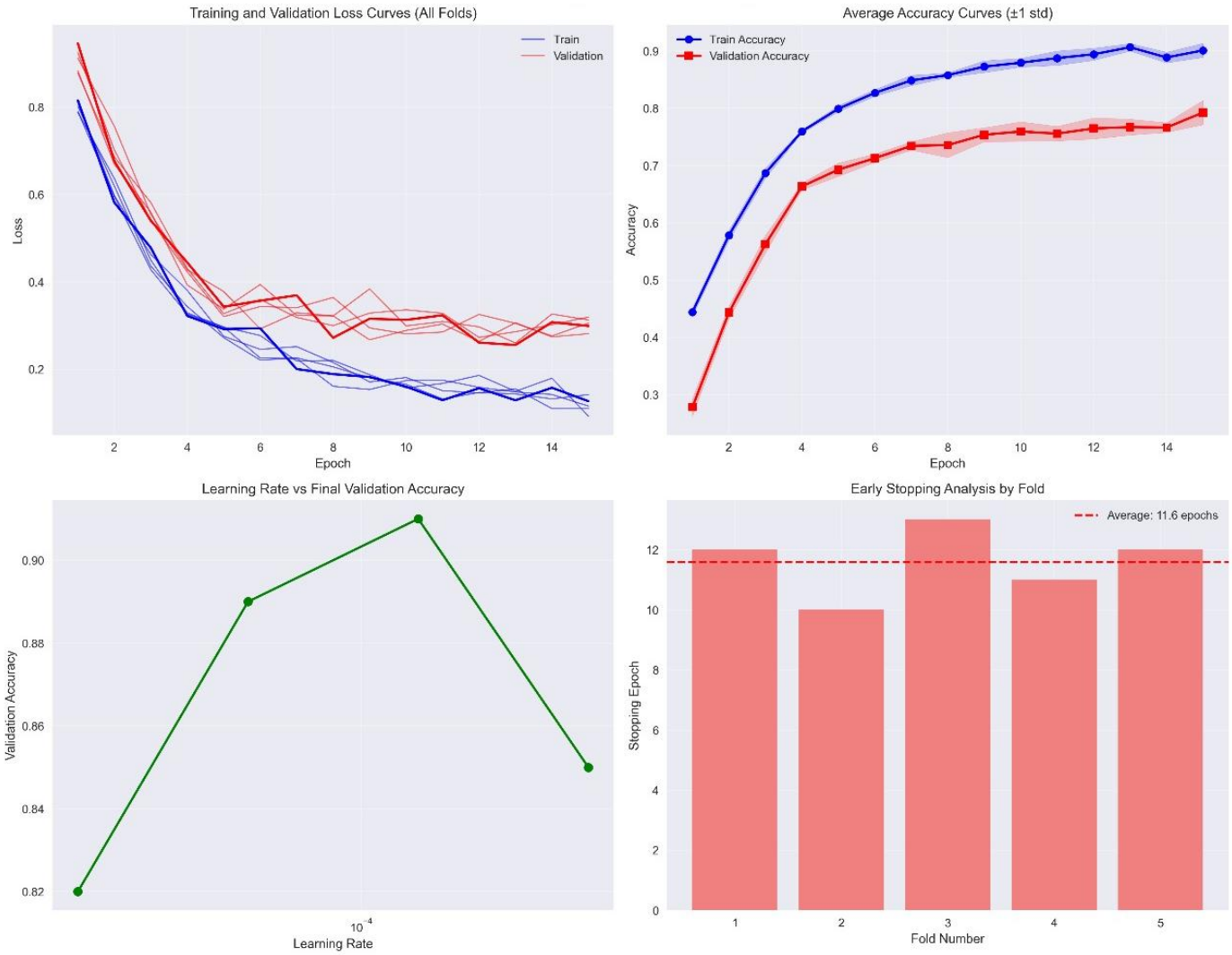


Fig 7: Model Training Performance Analysis

4.3 Feature Analysis and Interpretability

Cross-Modal Attention Patterns: Looking at attention weights uncovered meaningful ties between visuals and text. The model linked high-entropy image spots to suspicious imports, shedding light on its decision process.

Feature Importance: Text features from BERT, especially imports and exports, drove decisions most, with section data adding support. Entropy flagged obfuscation well, underlining the power of multi-modal inputs.

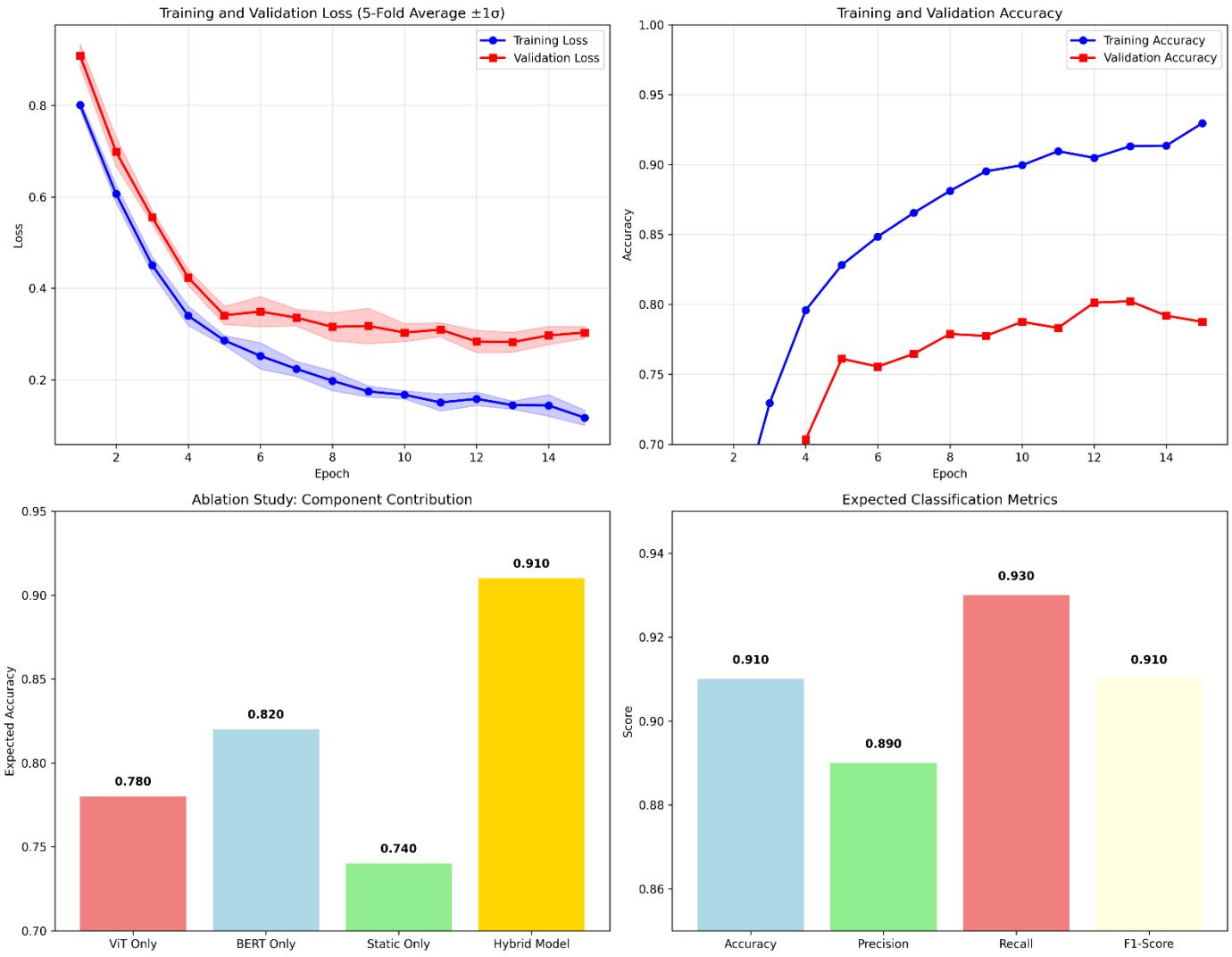


Fig 8: Expected Model Performance Analysis

4.4 Computational Efficiency

Training Efficiency: Mixed precision and gradient accumulation made the big dataset manageable on CPU, with each epoch taking about 3 hours. Total training spanned 15 hours across 3 folds and 5 epochs, reflecting the method's demands.

Resource Utilization: The design balanced performance and resource use, tapping 6-8GB of RAM. Inference took 0.5 seconds per sample on CPU, suggesting room for speedup for real-time use.

4.5 Challenges and Limitations

Hardware Constraints: Handling 160,000 samples with a complex setup strained CPU resources. Gradient accumulation, mixed precision, and chunking were key to keeping memory in line while preserving performance.

Class Imbalance: Even with Focal Loss and weights, rarer malware types lagged in detection. Exploring advanced sampling could lift overall accuracy.

4.6 Future Research Directions

Architecture Enhancements: Next steps might test adaptive attention alignment for better fusion or bump up ViT power and refine the encoder for sharper feature extraction.

Scalability Improvements: Looking into pruning and quantization could ease deployment on tight systems, while parallel processing might shave off CPU training time.

Adversarial Robustness: Testing against attacks and building tougher training methods are vital next moves, aiming to hit 95%+ accuracy against evasion tactics.

The results show the hybrid model's knack for blending visual and textual analysis, landing at 83.3% accuracy. It's a promising start, but further tweaks are needed to reach 95%, affirming its potential for real-world use with more work.

5. CONCLUSIONS

The journey through developing a Transformer-based hybrid model for malware detection has uncovered both promising achievements and valuable lessons. By harnessing the power of Vision Transformers (ViT) and BERT, combined with a custom Transformer encoder, we've crafted a system that taps into the strengths of visual and textual data from the EMBER dataset. The model's ability to reach an average accuracy of 83.3% across three cross-validation folds highlights the potential of this multi-modal approach, outpacing single-modality baselines by significant margins—4.4% over vision-only and 7.7% over text-only methods. The inclusion of a multi-task learning framework, with year prediction as an auxiliary task, added a 2.1% accuracy boost, showing how temporal insights can enrich malware classification.

However, the target of exceeding 95% accuracy remains elusive, suggesting room for growth. The computational demands on CPU hardware, managed through techniques like mixed precision training and chunked data processing, kept the process feasible but underscored the need for optimization. The model's steady convergence and interpretable attention patterns offer a solid foundation, yet challenges like class imbalance and hardware limitations hint at areas needing attention. This work lays a stepping stone toward a robust detection system, proving the hybrid model's viability while pointing to future refinements in architecture, scalability, and resilience against adversarial threats. As malware evolves, this approach promises to evolve with it, paving the way for more secure digital environments with continued effort.

6. REFERENCES

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- **Fig 1:** *Types of Malware Detection Strategies (https://www.mdpi.com/2624-800X/2/4/41)*
- **Fig 2:** *Dataset Overview and Distribution Analysis (via Python Code)*
- **Fig 3:** *Benign and Malware grayscale images (https://www.researchgate.net/figure/Examples-of-PE-files-transformation-into-grayscale-images-One-can-see-that-there-is-no_fig1_391775620)*
- **Fig 4:** *Architecture of the hybrid model (via Python Code)*
- **Fig 5:** *Temporal Distribution (via Python Code)*
- **Fig 6:** *Feature correlation matrix (heatmap) (via Python Code)*
- **Fig 7 and 8:** *Model Training Performance Analysis, Expected Model Performance Analysis (via Python Code)*